

Two global traces recover rational Erdős–Straus denominators

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Abstract

If $n \geq 3$ positive rational numbers have reciprocal sum $4/p$, their first $n - 1$ power sums are integers exactly when all the numbers are integers. For the three-denominator equation, two traces suffice. We prove a more general common-denominator law, then give a one-trace return and complete coefficient fibres on the original integer gate source. This is a global integrality test, not a proof that the relevant source is occupied at every prime.

Theorem 1. *Let $x_1, \dots, x_n \in \mathbb{Q}^\times$, $n \geq 2$, and suppose their power sums $s_k = \sum_i x_i^k$ are integers for $1 \leq k < n$. If their nonzero reciprocal sum is m/b in lowest terms, $b > 0$, then all reduced denominators are the same integer d , and*

$$\gcd(d, b) = 1, \quad d^n \mid (n - 1)!m.$$

If n is odd, d is odd.

Proof. For elementary symmetric functions e_j , Newton’s identities $je_j = \sum_{k=1}^j (-1)^{k-1} e_{j-k} s_k$ give $j!e_j \in \mathbb{Z}$ for $j < n$. They follow by taking the formal logarithmic derivative of $\prod_i (1 + x_i T)$; see Euler [1, §§3,5–7]. At a denominator prime ℓ , suppose the minimum root valuation is $-a < 0$ and exactly $r < n$ roots attain it. Their product is the unique least-valuation term of e_r , so $v_\ell(e_r) = -ar$. But $v_\ell(e_r) \geq -v_\ell(r!) > -r \geq -ar$, a contradiction. Hence every root has the same negative valuation at every denominator prime, proving the common reduced denominator assertion. Writing $x_i = A_i/d$ with each A_i a unit at primes of d , use $e_{n-1} = (m/b)e_n$ and $e_n = \prod_i A_i/d^n$. At $\ell^a \parallel d$ this gives

$$na \leq v_\ell((n - 1)!) + v_\ell(m) - v_\ell(b).$$

The factorial valuation is less than n , so $\ell \mid m$ and $\ell \nmid b$. The divisibility follows. If n were odd and d even, the scaled sum of its n odd reduced numerators would be odd, contradicting $s_1 \in \mathbb{Z}$. $\square$

Corollary 2. *For $n \geq 3$, $p \in \mathbb{Z}_{>0}$ and $x_i \in \mathbb{Q}_{>0}$ with $\sum_i 1/x_i = 4/p$,*

$$(x_1, \dots, x_n) \in \mathbb{Z}^n \iff (s_1, \dots, s_{n-1}) \in \mathbb{Z}^{n-1}.$$

In particular two integral traces suffice for the Erdős–Straus equation.

Proof. The reduced reciprocal numerator divides four, so Theorem 1 excludes odd denominator primes. Odd n excludes two. For even $n \geq 4$, $n - 1$ has at least two nonzero binary digits, so $v_2((n - 1)!) + 2 = n + 1 - s_2^{\text{bin}}(n - 1) \leq n - 1$; this also excludes two. The converse is immediate. For $n = 2$ the statement fails at $(p/2, p/2)$ with odd p . $\square$

The numerator matters: $(1/3, 1/3, 4/3)$ has first two traces $(2, 2)$ and reciprocal sum $27/4$. The full source edition further proves that at $n = 3$ the possible common denominator is exactly an odd d with $v_3(d) \leq 1$ and every other prime factor $1 \pmod{3}$, and constructs all such denominator types.

A single trace on the original integer gate

Let $p \equiv 1 \pmod{4}$ be prime, $p/4 < a < p/2$, $R = 4a - p$. Before imposing a divisor budget, take a positive integer U with $R \mid a + U$ and set

$$(x, y, z) = \left(a, \frac{pa(U + a)}{RU}, \frac{p(a + U)}{R} \right), \quad S = p + x + y + z.$$

This is a positive rational ES identity, with $z \in \mathbb{Z}$. Because R is a unit modulo U , its exact trace denominator is

$$\text{den } S = \frac{U}{\gcd(U, pa^2)}.$$

Indeed the integral numerator $pa(U + a)/R$, reduced modulo U , has the same gcd with U as pa^2 . Thus an integral trace gives exactly $U \mid pa^2$. At $v_p(U) = 0$ it returns the original M divisor $u = U$; at $v_p(U) = 1$ it returns the original E divisor $u = U/p$. The gate in the latter case becomes $4u + 1 = 0$ modulo R . These are the original Type-I/II coordinates [2, §2], not a new parametrization.

Explicitly, with $g = \gcd(a, u)$, set $h = g^2/u$, $r = u/g$, $s = a/g$. For M, $\lambda = (r + s)/R$ returns $(a, phs\lambda, phr\lambda)$; for E, $\kappa = (pr + s)/R$ returns $(a, hsk\kappa, phr\kappa)$. Their ordered inverses are $pa^2/(Ry - pa)$ and $a^2/(Ry - pa)$. The original exponent of each prime is recovered from u . An integral trace has tag M precisely when $S \equiv a \pmod{p}$.

Let $\Omega = \{1 \leq U \leq pa^2 : R \mid a + U\}$, $\Theta = \{S(U) : U \in \Omega\}$, and let $\Phi : \mathbb{Z}^\Omega \rightarrow \mathbb{Z}^\Theta$ send each original word to its trace label. The identity

$$S(U) - S(V) = \frac{p}{R}(U - V) \left(1 - \frac{a^2}{UV}\right)$$

proves that its only two-point fibres are the original middle complements $U, V = a^2/U$. The fixed point $U = a$ is excluded by $R \nmid 2a$. Exterior traces and nonintegral traces have singleton fibres. Hence, retaining both middle orientations,

$$\text{rank ker } \Phi = |M_a|/2, \quad |\Theta \cap \mathbb{Z}| = |E_a| + |M_a|/2.$$

The actual kernel basis is $e_U - e_{a^2/U}$ for $U < a$ in M. Original integer multiplicity two is not replaced by its zero reduction modulo two.

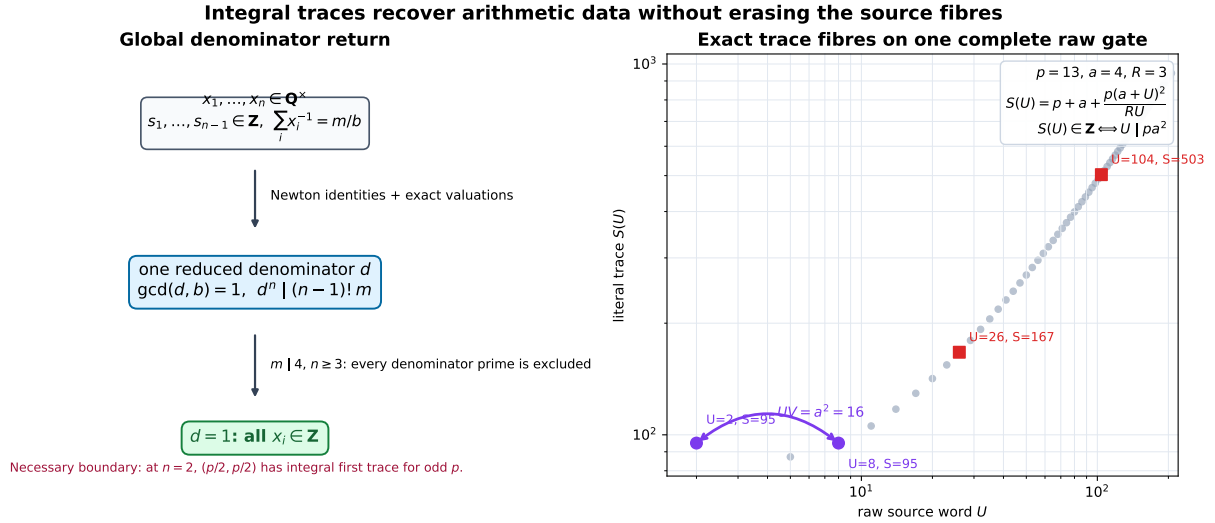


Figure 1: Left: the exact implication proved by the common-denominator theorem, including the necessary $n = 2$ boundary. Right: the complete raw gate at $p = 13$, $a = 4$, $R = 3$. The four integral returns are $U = 2, 8, 26, 104$; $U = 2$ and $U = 8$ form the unique two-point fibre because $UV = a^2 = 16$, while the two exterior returns are singletons. Every grey point is a retained raw word, so the diagram displays rather than removes the trace kernel. The script `figures/trace_integrity_fibres.py` computes every plotted value with exact rational arithmetic.

The first trace alone and the rationality hypothesis

Without the full integer gate, an integral trace can leave a square-part defect. For positive rational tails with $1/y + 1/z = R/(pa)$ and $N = y + z \in \mathbb{Z}$, the two reduced denominators are the same d . From $yz = Npa/R$ and $\gcd(pa, R) = 1$,

$$d^2 = R/\gcd(R, N), \quad \text{den}(a^2 + y^2 + z^2) = d^2.$$

The second equality uses odd R . Thus squarefree R suffices, but at the hard prime 1009 the exact rational identity

$$(x, y, z) = (286, 6413/3, 168238642/3)$$

has first trace 56081971 and second trace 28304240703866897/9.

Rational roots are essential in Corollary 2. Let b_1, b_2, b_3 be the roots of $T^3 - 6T^2 + 8T - 2$. Opposite signs at 0, 1, 2, 5 prove there are three positive roots. Reduction modulo three has no root, so the cubic is irreducible. The algebraic-integer denominators pb_i satisfy $\sum_i 1/(pb_i) = 4/p$ and have all power traces integral, beginning with $6p$ and $20p^2$, but none is rational. The full source edition constructs a stronger such family at every hard prime, with an integer first denominator and the original middle prime-local cluster signature. It is outside the rational-source domain, not an ES counterexample.

The remaining assertion is that some original shell has an integral trace in Ω (equivalently an original E or M divisor hit). No uniform positive lower bound on that count or kernel has been proved here. The work is an exact global integrality return, not universal source occupancy or a completed Turn 7.

References

- [1] Euler, L. (1750/2007). *A double demonstration of a theorem of Newton, which gives a relation between the coefficient of an algebraic equation and the sums of the powers of its roots* (J. Bell, Trans.). E153; arXiv:0707.0699, §§3,5–7. Classical Newton identities; the proof above is self-contained.
- [2] Elsholtz, C., & Tao, T. (2013). Counting the number of solutions to the Erdős–Straus equation on unit fractions. *Journal of the Australian Mathematical Society*, 94, 50–105. arXiv:1107.1010, §2, Propositions 2.2 and 2.6.